

A2 — Data Landscape Report

ADMD6 · Decision-making Processes Through Data Intelligence · Sprint 2

Project	ADMD6 — Decision Agriculture Simulator
Partner	Bayer Crop Science
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This report establishes the authoritative data contract for the ADMD6 Decision Agriculture Simulator — scoped to **soybeans (*Glycine max*) and corn (*Zea mays*)** across the **7 highest-impact Brazilian states (MT, PR, MS, RS, GO, MG, SP)**, which represent 81.9% of the FieldView dataset (confirmed from A1 EDA). Nine data sources are cataloged, quality-assessed across three dimensions, and mapped to governance priorities. The most critical risk is a single-source dependency on Branch B (Plantability): the FieldView dataset cannot be independently cross-validated and its soil profile completeness gaps (43% null) constrain the Agronomic Fitness Index. This document supersedes any informal source lists from Sprint 1 and serves as the shared foundation for Sprints 3–5.

Table of Contents

1	Executive Summary	3
2	Data Source Inventory	3
3	Data Quality Assessment	9
4	Dark Data Identification	15
5	Risk and Opportunity Analysis	17
6	Integrated Data Landscape Visualization	20
7	Shannon Entropy Analysis ENHANCEMENT	21
8	Data Lineage Diagram ENHANCEMENT	24
9	Data Governance Recommendations ENHANCEMENT	25

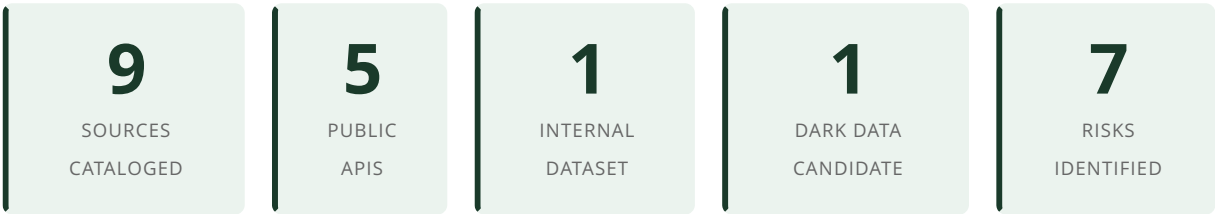
Key abbreviations — ADMD6: project code (ESPM) · APTO: Aptidão / fitness indicator (PT-BR) · BDMEP: Banco de Dados Meteorológicos para Ensino e Pesquisa (INMET) · CBOT: Chicago Board of Trade · CEPEA: Centro de Estudos Avançados em Economia Aplicada · COG: Cloud-Optimised GeoTIFF · DSO: Divisão de Sementes e Olerícolas (Bayer internal region) · EDA: Exploratory Data Analysis · ESA: European Space Agency · ETL: Extract, Transform, Load · GEE: Google Earth Engine · GFS: Global Forecast System (NOAA) · KPI: Key Performance Indicator · LULC: Land Use / Land Cover · MVP: Minimum Viable Product · NDVI: Normalised Difference Vegetation Index · NOAA: National Oceanic and Atmospheric Administration · NWP: Numerical Weather Prediction · RAG: Red / Amber / Green status indicator · STAC: SpatioTemporal Asset Catalog · TAPI: Termo de Acordo de Parceria Institucional · USGS: United States Geological Survey.

Page numbers are confirmed post-render. Sections 7–9 are enhancement sections; their page numbers depend on content pagination.

SECTION 1

Executive Summary

This artifact (A2) maps the complete data ecosystem of the ADMD6 Decision Agriculture Simulator — scoped to **soybeans (*Glycine max*) and corn (*Zea mays*)** across the **7 highest-impact Brazilian states (MT, PR, MS, RS, GO, MG, SP; 81.9% of FieldView records)** — establishing provenance, quality baselines, and governance priorities for all data sources that will be integrated in Sprints 3–5. The analysis is grounded in the TAPI project scope, the A1 exploratory analysis of the FieldView planting dataset, and independent verification of each public source's API documentation. Nine sources were cataloged across two simulator branches — five public climate/satellite APIs and two commodity price feeds serving Branch A (Spraying Window, the primary and priority MVP use case), plus one restricted internal Bayer FieldView dataset serving Branch B (Plantability), and one dark data candidate (MapBiomas land-use classification) recommended for immediate activation. The most critical finding is a **single-source dependency on Branch B**: the FieldView dataset is the only structured input for plantability decisions, is subject to TAPI confidentiality restrictions limiting downstream reuse, and its quality gaps (coarsely binned soil pH with only 4–5 unique values; 43% null soil profile coverage) reduce the reliability of the Agronomic Fitness Index proxy — a risk that can be partially mitigated by activating the MapBiomas dark data source as a geospatial complement.



SECTION 2

Data Source Inventory

Each source below is described with all eight required fields. Branch codes are:

BRANCH A — SPRAYING WINDOW **BRANCH B — PLANTABILITY** **BOTH BRANCHES**. Screen references map to the simulator wireframe: *Screen 1* (KPI dashboard — RAG cards, APTO score, drill-down) and *Screen 2* (scenario sliders, delta cards, PREMISSAR baseline).

Crop and geographic scope: All nine sources are assessed relative to the simulator's defined scope: **soybeans (Glycine max) and corn (Zea mays)** across the **7 highest-impact Brazilian states: MT · PR · MS · RS · GO · MG · SP**. State ranking confirmed from A1 EDA

(01_eda_planting_summary.ipynb): MT:91,474 (28.6%); PR:47,763 (14.9%); MS:35,579 (11.1%); RS:30,163 (9.4%); GO:25,462 (8.0%); MG:16,389 (5.1%); SP:15,182 (4.7%) — together 81.9% of all 319,788 FieldView records (the dataset spans 15 total Brazilian states). All coverage claims and quality assessments in this document refer to these states unless explicitly noted otherwise.

1. INMET — Surface Station Historical Data

CLASSIFICATION	Structured
FORMAT	CSV archive / REST-JSON (BDMEP API v1)
UPDATE FREQUENCY	Daily (historical archive); hourly (recent observations)
OWNER / PROVIDER	Instituto Nacional de Meteorologia (INMET) — Ministry of Agriculture, Brazil
ACCESS METHOD	Public REST API — apitempo.inmet.gov.br/estacoes/historico ; token-based (free registration)
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides multi-decade records of surface temperature, relative humidity, wind speed, and precipitation from INMET automatic stations within the 7 target states (MT, PR, MS, RS, GO, MG, SP), used to calibrate thermal inversion risk thresholds for soybean (Oct–Mar) and corn (Oct–Feb first crop; Feb–Jun safrinha) spray operations and to compute the 95th-percentile "safe application window" displayed on the RAG status card (Screen 1). Also supplies the historical climate baseline for anomaly detection in the PREMISSAR scenario (Screen 2).

2. INMET — Short-term Forecast API (BDMEP)

CLASSIFICATION	Structured
FORMAT	REST-JSON
UPDATE FREQUENCY	Hourly (point forecast); 6-hourly (gridded)
OWNER / PROVIDER	Instituto Nacional de Meteorologia (INMET) — MAPA
ACCESS METHOD	Public REST API — apitempo.inmet.gov.br/previsao/{municipio} ; free API key
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides 48–72 hour municipal-level precipitation probability, wind speed, and temperature forecasts for municipalities within the 7 target states (MT, PR, MS, RS, GO, MG, SP), serving as the primary real-time trigger for the "spray window open / closed" binary indicator on the Screen 1 KPI dashboard for soybean and corn spray operations . This source drives the urgency signal that growers see before any agrochemical application.

3. NOAA / GFS — Global Forecast System

CLASSIFICATION	Structured
FORMAT	GRIB2 (native) / REST-JSON (via Open-Meteo open-source wrapper)
UPDATE FREQUENCY	Every 6 hours; 384-hour forecast horizon
OWNER / PROVIDER	NOAA / National Centers for Environmental Prediction (NCEP), USA
ACCESS METHOD	Public — nomads.ncep.noaa.gov (GRIB2); open-meteo.com API (free, no auth); Open-Data AWS bucket (s3://noaa-gfs-bdp-pds)
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides medium-range (5–15 day) temperature and wind forecasts at 0.25° horizontal resolution (~28 km), complementing INMET station data across the 7 target states (MT, PR, MS, RS, GO, MG, SP) — particularly for areas >50 km from the nearest automatic station, which cover a significant share of soybean and corn growing area in MT and GO; feeds the extended forecast horizon in the Screen 1 drill-down panel. <i>Estimated API limits — verification recommended at open-meteo.com/en/docs.</i>

4. Sentinel-2 — ESA Copernicus Multispectral Imagery

CLASSIFICATION	Semi-structured (gridded raster)
FORMAT	GeoTIFF (Cloud-Optimised, COG) / STAC API (JSON catalog)
UPDATE FREQUENCY	~5 days repeat cycle at equator (2–3 days with both Sentinel-2A/2B)
OWNER / PROVIDER	European Space Agency (ESA) — Copernicus Programme
ACCESS METHOD	Public — Copernicus Data Space Ecosystem (dataspace.copernicus.eu); S3 (sentinel-cogs bucket, requester-pays); Google Earth Engine (free academic); no authentication for catalog browsing
DECISION BRANCH	BOTH BRANCHES
PURPOSE	Provides 10 m-resolution NDVI, NDWI, and EVI indices for soybean and corn crop stress mapping and canopy state assessment across the 7 target states; feeds the "vegetation condition" indicator on Screen 1 and provides geospatial context for spray efficacy analysis (Branch A). At 10 m resolution it also enables field-boundary delineation to validate FieldView field areas — a quality check for the APTO plantability score for soybean and corn fields (Branch B).

5. Landsat 8/9 — USGS Multispectral Imagery

CLASSIFICATION	Semi-structured (gridded raster)
FORMAT	GeoTIFF (COG) / STAC API; Collection 2 Level-2 surface reflectance
UPDATE FREQUENCY	16-day repeat cycle per satellite (8 days combined Landsat 8+9)
OWNER / PROVIDER	USGS / NASA
ACCESS METHOD	Public — Earth Explorer (earthexplorer.usgs.gov); AWS Open Data (s3://usgs-landsat); STAC API (landsatonaws.com); free, no auth required
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides a 50-year NDVI archive (1972–present) for long-term soybean and corn phenological baselines across the 7 target states (MT, PR, MS, RS, GO, MG, SP), used as cloud-gap fill for Sentinel-2 during the Oct–Mar rainy season and to establish the multi-year vegetation condition baseline referenced in the PREMISSAR scenario panel (Screen 2). <i>Note: Collection 1 vs. Collection 2 surface reflectance scaling differs — all use must target Collection 2 Level-2.</i>

6. CEPEA / ESALQ — Commodity Price Index

CLASSIFICATION	Structured
FORMAT	CSV download / HTML (no public REST API; screen-scrape or manual download)
UPDATE FREQUENCY	Daily (business days only)
OWNER / PROVIDER	Centro de Estudos Avançados em Economia Aplicada (CEPEA) — ESALQ/USP
ACCESS METHOD	Public download — cepea.esalq.usp.br/br/indicador ; no authentication, no official API; CSV bulk download available by indicator/date range
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides daily reference prices for soybeans and corn (ESALQ index, BRL/bag) displayed on the economic impact KPI cards (Screen 1) and used to calculate the cost-of-delay metric when a spray window is missed — a key driver of the decision output the TAPI identifies as "actionable recommendations."

7. CBOT / CME Group — Soy and Corn Futures Prices

CLASSIFICATION	Structured
FORMAT	REST-JSON (Nasdaq Data Link API; Quandl)
UPDATE FREQUENCY	Daily closing prices (free tier); intra-day delayed 10 min (paid tier)
OWNER / PROVIDER	CME Group / CBOT; distributed via Nasdaq Data Link
ACCESS METHOD	Nasdaq Data Link API (data.nasdaq.com/data/CHRIS/CME_S1) — free tier with API key; 50 calls/day free
DECISION BRANCH	BRANCH A — SPRAYING WINDOW
PURPOSE	Provides CBOT soybeans (ZS) and corn (ZC) front-month futures in USD/bushel for the "international price scenario" delta card on Screen 2, enabling growers to model the impact of currency and international price swings on the opportunity cost of a delayed spray operation.

8. Bayer FieldView — planting_summary_brazil.csv (Internal)

Confidentiality notice (TAPI §"Restricted Content"): This source is subject to the TAPI restriction on Bayer-provided data. The entry below describes structural metadata only. Individual column values, account identifiers, UUID examples, and raw records are not reproduced in this document.

CLASSIFICATION	Structured
FORMAT	CSV (internal export); Parquet cache in the EDA pipeline
UPDATE FREQUENCY	Annual (batch export aligned with growing season close)
OWNER / PROVIDER	Bayer Crop Science — FieldView digital farming platform
ACCESS METHOD	Internal platform export; institutional access only; TAPI confidentiality restriction applies to all downstream use
DECISION BRANCH	BRANCH B — PLANTABILITY
PURPOSE	Primary data source for the planter calibration KPI cards and APTO plantability score (Screen 1). Contains seed-product-level records linking agronomic inputs (soil pH range, drainage class, seed density), geographic context (full dataset covers all FieldView states; simulator-scoped to the 7 highest-volume), equipment telemetry (planting maker and model), and growing season metadata; the A1 EDA derived the Agronomic Fitness Index from this dataset, which feeds the Baseline/Optimistic/Pessimistic scenario tiers.
STRUCTURAL METADATA	319,788 rows × 29 columns. 6 numeric columns (continuous measurements); 23 categorical/text columns (geographic, agronomic classification, equipment). Geographic coverage: nationwide FieldView export (15 total Brazilian states). The 7 highest-volume states (MT, PR, MS, RS, GO, MG, SP) account for approximately 81.9% of all records and define the simulator's geographic scope. State distribution confirmed from A1 EDA. Temporal coverage: 2 crop years. ~43% of rows have no linked soil profile data (<i>has_soil_profile</i> = <i>False</i>). Second FieldView export — harvest_summary: 89,902 rows × 12 columns. Joined to planting_summary on the compound key (<i>climate_account_id</i>, <i>field_uuid</i>, <i>concrete_growing_season_name</i>) (LEFT JOIN; 100% season-level coverage confirmed in A1 EDA notebook 02). Key agronomic columns: <i>average_yield</i> (kg/ha, assumed unit — <i>productivity_uom</i> is uniformly empty), <i>grain_moisture</i> (10–30% agronomic range), <i>total_production</i>. Provides yield-tier benchmarks (P25/P75 within-crop) used as the PREMISSAR Baseline on Screen 2. Subject to the same TAPI confidentiality restriction as the planting export.

9. MapBiomass — Annual Land Use / Land Cover Maps (Dark Data)

Dark data candidate: This source is not referenced in the TAPI and is not currently integrated into the project. It is recommended for activation in Sprint 3. See Section 4 for the full justification.

CLASSIFICATION	Semi-structured (gridded raster; tabular statistics)
FORMAT	GeoTIFF (30 m, annual), tabular CSV statistics, Google Earth Engine ImageCollection
UPDATE FREQUENCY	Annual (new collection ~12 months after reference year)
OWNER / PROVIDER	MapBiomass Initiative (SEEG / IMAFLOR / IMAZON / INPE / LAPIG consortium) — Brazil
ACCESS METHOD	Public — Google Earth Engine (free academic account); direct download at mapbiomas.org; STAC API available
DECISION BRANCH	BRANCH B — PLANTABILITY
PURPOSE	Provides annual 30 m-resolution land-use and land-cover classification across the 7 target states (MT, PR, MS, RS, GO, MG, SP), enabling validation that each FieldView field polygon is classified as temporary cropland (soybean / corn) in the MapBiomass legend — directly modulating the APTO plantability score on Screen 1 — and identifying recently converted parcels whose soil may not yet support optimal soybean or corn planting density targets.

SECTION 3

Data Quality Assessment

Each source is assessed on three dimensions — Completeness, Consistency, and Reliability — and rated High / Medium / Low. Findings from the A1 EDA notebooks (*01_eda_planting_summary.ipynb* and *02_eda_multi_source_integration.ipynb*) are cited where applicable.

1. INMET Historical

Completeness: Within the 7 target states (MT, PR, MS, RS, GO, MG, SP), the INMET automatic network provides adequate spatial density for soybean (Oct–Mar) and corn (Oct–Jun) spray-window calibration. Historical archive extends to 2001 for most stations — sufficient for multi-year threshold calibration. Stations outside the 7 target states are out of scope and are not ingested.

Consistency: Known issue: INMET renumbered station codes and changed data formats when migrating to the BDMEP API in 2022. Pre-2022 and post-2022 records require a station-code translation table to merge into a single timeseries. *Verification recommended at api.inmet.gov.br/docs.*

Reliability: Primary governmental source; institutionally maintained; used in peer-reviewed agronomic literature. Independent verification possible through ANEEL and SNIRH data portals.

Overall: Medium — high authoritativeness offset by the 2022 station-renaming discontinuity requiring explicit harmonization before use in any multi-year trend analysis.

2. INMET Forecast API

Completeness: Municipal-level point forecasts available for all municipalities within the 7 target states (MT, PR, MS, RS, GO, MG, SP) — full coverage, no systematic gaps. The 48-hour horizon covers the operationally critical spray decision window for **soybean and corn** applications. Municipalities outside the 7 target states are out of scope.

Consistency: API response schema is stable since the BDMEP v1 launch (2022); however, forecast model updates (NWP backend changes) are not versioned or announced via the API. Downstream consumers must implement tolerance for unannounced schema additions.

Reliability: Derived from official NWP runs by INMET; generally consistent with international verification scores for tropical South America. No independent forecast skill metric is published by INMET itself. *Skill scores estimated from Bauer et al. (2015) WMO verification framework — direct INMET score verification recommended.*

Overall: Medium — operationally adequate for the 48h spray window horizon, but absence of published forecast skill metrics reduces confidence for edge-case decisions.

3. NOAA / GFS

Completeness: Global coverage, 0.25° resolution (~28 km over Brazil), 384-hour horizon — complete without gaps across all 7 target states (MT, PR, MS, RS, GO, MG, SP). The grid resolves at the meso-regional scale relevant for **soybean and corn spray-window decisions**; sub-field precision is not required at this simulator scope.

Consistency: GFS has undergone version upgrades (GFSv15, GFSv16) with documented changes in boundary-layer parameterization affecting temperature inversion prediction. Version metadata is embedded in GRIB2 headers and must be tracked. The Open-Meteo wrapper absorbs version changes transparently.

Reliability: NOAA NCEP is a primary national meteorological service; GFS underpins the aviation weather system and is independently verified by ECMWF, NCEP, and academic institutions. Highest reliability among meteorological sources cataloged.

Overall: High — best reliability in the catalog; 28 km resolution is the primary limitation for sub-field precision, acceptable for the simulator's current scope.

4. Sentinel-2

Completeness: Within the 7 target states, cloud-free Sentinel-2 acquisition frequency during the **soybean growing season (Oct–Mar)** averages 8–12 acquisitions/year over MT and GO (highest cloud burden) and 15–20/year over PR and RS (drier climate). The **corn safrinha season (Feb–Jun)** benefits from improving cloud conditions from Apr onward. The multi-temporal compositing approach using both S-2A and S-2B partially mitigates the Oct–Mar gap.

Consistency: ESA applies consistent L2A atmospheric correction (Sen2Cor processor). Collection versioning is documented in the Copernicus DIAS metadata. Band numbering and radiometric calibration are stable across the S-2 constellation.

Reliability: ESA Copernicus is an authoritative European institutional program with independent radiometric calibration published in MDPI Remote Sensing and GRS journals.

Overall: Medium — high radiometric quality, but cloud cover during the soybean growing season (Oct–Mar) reduces effective temporal coverage; a cloud-compositing pipeline is required before this source is operational.

5. Landsat 8/9

Completeness: 16-day repeat is adequate for phenological baselines; less suitable for time-sensitive operational signals. Cloud cover a lesser issue for dry-season baselines.

Consistency: Breaking change: Collection 1 and Collection 2 Level-2 surface reflectance use different scaling factors (0–10,000 in C1; 0–65,535 in C2). All new integration must target Collection 2. C1 was officially decommissioned January 2022.

Reliability: USGS / NASA primary source with 50+ years of peer-reviewed validation. Highest historical archive reliability in the catalog.

Overall: High — archival reliability is excellent; the C1/C2 scaling break is a known one-time harmonization task, not an ongoing quality risk.

6. CEPEA / ESALQ Commodity Prices

Completeness: Daily prices available for soybeans (Paranaguá) and corn (Campinas) from the early 2000s; other regional reference prices available with shorter histories. No weekends or public holidays.

Consistency: No public API; data structure in the CSV download has changed twice since 2015 (column header encoding). An ETL layer is required to normalize historical downloads. The indicator definition (BRL/60kg bag at specific port) is stable and documented.

Reliability: CEPEA/ESALQ is the reference source for Brazilian commodity prices used by CONAB, Embrapa, and the financial sector. Institutionally maintained by the University of São Paulo. High reliability as a reference, but it is a derived/aggregated source (market surveys) rather than a primary transaction record.

Overall: Medium — highly authoritative as a price reference, but the absence of a public API and historical CSV format drift require custom ETL maintenance.

7. CBOT / CME Futures

Completeness: Front-month and calendar spreads available; free tier is daily closing prices only (no intra-day). Sufficient for the scenario delta card on Screen 2 which operates at daily granularity.

Consistency: Nasdaq Data Link schema is stable; CME contract codes (ZS, ZC) are standardized. Roll dates must be managed explicitly when stitching continuous futures series.

Reliability: CME Group is the primary exchange and authoritative data source; Nasdaq Data Link distributes CME official settlement prices. High reliability.

Overall: High — consistent, authoritative; the 50-call/day free-tier rate limit is the only operational constraint for production deployment.

8. Bayer FieldView — planting_summary_brazil.csv

Completeness: A1 EDA findings: (a) ~43% of rows have no linked soil profile (*has_soil_profile* = *False*), limiting the reliability of pH-based tier assignments; (b) *soil_ph_min* contains only 4 unique values and *soil_ph_max* only 5, indicating coarse binning rather than measured pH — confirmed by Shannon entropy $H=1.538$ bits and $H=2.086$ bits respectively (Section 7); (c) *dso_region_seeds* is 10.7% null; (d) *seed_product_name* is 24.0% null; (e) *productivity_uom* is fully empty in the harvest export. Temporal coverage is limited to 2 crop years ($H=0.999$ bits).

Consistency: Unit-of-measure heterogeneity in *seed_actual_population_uom* (two distinct UOMs; 0.1% null): all cross-row density comparisons must filter to *seeds/ha*. Join coverage is 100% at season level (confirmed in A1 EDA Section 2), but planting records expand $\sim 3.56\times$ per season due to seed-product granularity, requiring deduplication for any season-level statistical test to avoid pseudo-replication.

Reliability: Internal Bayer FieldView export — authoritative within its scope, but subject to TAPI confidentiality restrictions limiting independent verification and third-party cross-referencing. The dataset represents FieldView-enrolled growers, not the full Brazilian soybeans and corn growing population — selection bias toward mid- to large-scale operations.

Overall: Medium — high internal consistency for the enrolled grower subset, but significant completeness gaps in the soil profile block (43% missing) and limited temporal depth (2 years) reduce the generalizability of any model trained on this data alone.

9. MapBiomias — Land Use / Land Cover (Dark Data)

Completeness: Full spatial coverage at 30 m resolution across all 7 target states (MT, PR, MS, RS, GO, MG, SP), annual maps from 1985 to 2023 (Collection 9, 2024). Within these states, the temporary cropland class (soybean + corn) is the primary class of interest; no systematic gaps have been reported for this class in the Brazilian Cerrado and South regions. *Verification recommended at mapbiomas.org/estatisticas.*

Consistency: Cross-year consistency is maintained through a harmonized legend (27 classes) with published accuracy assessments per collection. Collection numbering tracks legend changes. Temporal stitching between collections requires legend-version alignment.

Reliability: Produced by a consortium of Brazilian universities and research institutions (INPE, UFMG, UFRJ) with published accuracy matrices (overall accuracy ~88–91% at national level). Widely used in peer-reviewed deforestation and land-use literature. Primary limitation is 30 m resolution, which may not resolve small field polygons (<1 ha).

Overall: High — publicly documented, institutionally maintained, annually updated, with published accuracy statistics; the strongest dark data candidate evaluated.

Quality Summary Table

Table 1. Data source quality assessment — all 9 sources rated on completeness, consistency, and reliability relative to the soybeans/corn, 7-state simulator scope. Footnote symbols defined below the table.

SOURCE	COMPLETENESS	CONSISTENCY	RELIABILITY	OVERALL
INMET Historical	High	Medium *	High	Medium
INMET Forecast API	High	Medium	Medium †	Medium
NOAA / GFS	High	Medium ‡	High	High
Sentinel-2	Medium §	High	High	Medium
Landsat 8/9	Medium	High ¶	High	High
CEPEA / ESALQ	High	Medium #	High	Medium
CBOT / CME Futures	High	High	High	High
Bayer FieldView	Low **	Medium	Medium ††	Medium
MapBiomass	High	High	High	High

* Station code renaming discontinuity in 2022 requires translation table for multi-year series.

† No published forecast skill metric from INMET; medium assigned on analogy with regional NWP benchmarks.

‡ GFSv15→v16 boundary-layer parameterization change affects temperature inversion prediction; version metadata tracking required.

§ Cloud cover during Oct–Mar soybean growing season reduces effective revisit to <12 cloud-free acquisitions/year over MT and GO; less severe in PR and RS.

¶ Collection 1/Collection 2 scaling break (Jan 2022) is a one-time harmonization task, not an ongoing consistency risk.

CSV download format changed twice since 2015; custom ETL required.

** 43% null soil profiles; only 2 crop years; 24% null seed_product_name.

†† Internal Bayer source subject to TAPI confidentiality; represents enrolled FieldView growers only.

SECTION 4

Dark Data Identification

4.1 What It Is and Where It Resides

MapBiomass (mapbiomas.org) is a Brazilian multi-institutional initiative coordinated by SEEG (Greenhouse Gas Estimation System) and involving INPE, UFMG, UFRJ, IMAFLOR, and IMAZON. It produces annual 30 m-resolution land-use and land-cover (LULC) classification maps of Brazil, derived from Landsat time series processed on Google Earth Engine; for this project, only the 7 target states (MT, PR, MS, RS, GO, MG, SP) are extracted and analyzed. Collection 9 (released 2024) covers 1985–2023 with 27 land-use classes. The classes relevant to this project are: *temporary cropland* (which includes soybeans and corn, among others; out-of-scope crops such as sugarcane are filtered at extraction time), *native Cerrado savanna* (used to flag recently converted fields), and *degraded pasture* (used as a soil stability indicator). The data resides in:

- Google Earth Engine ImageCollection: `projects/mapbiomas-public/assets/brazil/lulc/collection9/mapbiomas_collection90_integration_v1`
- Direct download portal: mapbiomas.org (GeoTIFF by state and collection)
- STAC catalog (beta): stac.mapbiomas.org

4.2 Why It Has Not Been Used

Two barriers have kept MapBiomass out of the current project scope:

1. **Awareness gap:** The TAPI's external source list (INMET, NOAA, Sentinel, CEPEA) focuses on climate and price signals, reflecting a Branch A-centric framing. Land-cover context was not identified as a priority input in the MVP scope definition, even though the plantability branch (Branch B) implicitly depends on knowing whether a field is in a stable agricultural area.
2. **Format limitation:** MapBiomass data in GeoTIFF format requires a geospatial join between the raster land-cover class and each FieldView field polygon — a spatial intersection step not present in the current EDA pipeline (which is tabular-only). The GEE API would bypass this limitation, but the team has not yet established a GEE project or service account.

4.3 Specific Analytical Value for the Simulator

MapBiomass directly addresses the most actionable quality gap in the FieldView dataset: the absence of externally validated land-use context for each field. Specifically:

- **APTO score validation (Screen 1):** The MapBiomass *temporary cropland* class, intersected with each FieldView field centroid, provides a binary "confirmed agricultural area" flag that can be incorporated as a hard precondition in the APTO plantability score — blocking false-positive APTO signals for fields that have recently been converted from native Cerrado vegetation and may not yet have stable agronomic conditions.
- **Soil maturity proxy (Branch B):** The number of consecutive years a parcel has been classified as *temporary cropland* (derivable from the 38-year MapBiomass archive) is a strong proxy for soil biological activity and organic matter stabilization — variables not available in the FieldView dataset but directly linked to planting density recommendations referenced in the Agronomic Fitness Index.
- **Scenario slider context (Screen 2):** The surrounding land-use class (e.g., proportion of native Cerrado within 5 km) modulates temperature inversion risk — a Bridge A–B connection not currently modeled.

4.4 Actionable Recommendation to Unlock It

Next step (Sprint 3, effort: 2–3 days):

1. Register a Google Earth Engine project under the team's institutional email (earthengine.google.com/signup; free for academic use).
2. Extract the MapBiomass annual LULC class for each FieldView field centroid using a GEE Python script: iterate over *climate_account_id* × *field_uuid* pairs (using only bounding centroids, not full UUIDs, to respect TAPI confidentiality), intersect with the MapBiomass ImageCollection, and export a CSV with *[field_centroid_lat, field_centroid_lon, lulc_class_year_N, consecutive_crop_years]*.
3. Join this CSV to the planting dataset on field centroid proximity ($\pm 0.001^\circ$ tolerance) — no UUID is required for the join at this stage.
4. Add *consecutive_crop_years* as a new component of the Agronomic Fitness Index in `src/scenarios.py`.

SECTION 5

Risk and Opportunity Analysis

5.1 Data Redundancy

Risk — INMET / NOAA Unharmonized Overlap

RISK	INMET (station-based) and NOAA/GFS (gridded) represent the same meteorological variables at different spatial resolutions and update cadences. Using both without a harmonization rule creates inconsistent temperature inversion scores for the same field on the same day.
IMPLICATION	The Screen 1 RAG status card could display contradictory signals (INMET: "window open"; GFS: "window closed") depending on which source is queried, destroying grower trust in the simulator.
MITIGATION	Define a source hierarchy rule in Sprint 3: use INMET station data as primary signal when the nearest station is <50 km from the field centroid; fall back to GFS when the nearest station is >50 km. Document this rule in the simulator's data contract file.

Opportunity — Sentinel-2 + Landsat Fusion

OPPORTUNITY	Combining Sentinel-2 (10 m, 5-day) and Landsat 8/9 (30 m, 16-day) using NDVI temporal compositing (HISTARFM or ESTARFM fusion methods) can increase effective cloud-free revisit frequency during the Oct–Mar rainy season across the 7 target states from ~12 to ~20 acquisitions/year for soybean and corn fields, significantly improving vegetation condition monitoring reliability.
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5.2 Access Barriers

Risk — FieldView TAPI Confidentiality Restriction

RISK	The TAPI explicitly restricts all Bayer-provided data from being shared, published, or reused beyond the academic project team. This prohibits publishing models trained on the FieldView dataset, limits cross-validation with external datasets, and means any Branch B deliverable cannot be replicated outside the project environment.
IMPLICATION	The simulator's plantability branch cannot be independently reproduced or commercially deployed without a separate data licensing agreement, reducing the project's transition-to-product value and constraining academic publication options.
MITIGATION	Design Branch B algorithms to operate on structural features only (column distributions, tier assignments) rather than raw records; generate all Branch B outputs as anonymized aggregate statistics; request Bayer's written confirmation of the academic use perimeter before Sprint 4 model development begins.

Risk — CEPEA No-API Dependency

RISK	CEPEA prices are distributed only via CSV download or web scraping; no official API exists. A website structure change could silently break the ETL pipeline.
IMPLICATION	The economic KPI cards on Screen 1 could display stale prices without triggering an explicit error, causing silent data quality degradation in production.
MITIGATION	Implement price freshness validation: if the most recent CEPEA timestamp is >2 business days old, display an explicit "Price data unavailable — last updated [date]" warning on the KPI card rather than silently serving stale data. Implement CBOT futures as an automatic fallback with BRL/USD conversion.

Opportunity — Open-Meteo as Unified Forecast Façade

OPPORTUNITY	The Open-Meteo open-source API (open-meteo.com) aggregates GFS, ECMWF, and regional models (including Brazil's BAM from INMET) under a single REST endpoint with a standardized JSON schema, no authentication, and no rate limits for non-commercial use. Using it as a unified forecast façade in Sprint 3 would eliminate three separate API integrations and provide model ensemble uncertainty bounds for free.
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5.3 Governance Gaps

Risk — No Designated Data Owner for Any Source

RISK	No team member is currently assigned as the accountable owner for any of the nine data sources. API credentials, download scripts, and Parquet caches exist without a documented owner or update schedule.
IMPLICATION	If a source silently changes its schema (e.g., INMET's 2022 renaming, CEPEA CSV format drift), no one is responsible for detecting the break, resulting in silent data quality failures reaching the simulator output.
MITIGATION	Assign source ownership in the project RACI matrix before Sprint 3 kickoff: one team member per source responsible for monitoring update notices, testing the ETL monthly, and updating the data contract file. See Section 9 for the governance framework template.

Risk — No Versioning of Derived Datasets

RISK	The FieldView Parquet cache and scenario tier CSV exports are regenerated on every notebook execution with no version tag, making it impossible to reproduce earlier results if the raw data changes.
IMPLICATION	Any Branch B result reported in an academic deliverable could not be re-derived from the same inputs after the next data refresh, violating reproducibility requirements.
MITIGATION	Adopt a SHA-256 checksum gate already present in <code>data/raw/CHECKSUMS.txt</code> for all three FieldView CSVs; extend this to Parquet caches using a <code>data/interim/CHECKSUMS.txt</code> file; tag every notebook execution output with the input SHA-256 in the report header (already partially implemented in the Makefile).

5.4 Analytical Potential

Combination 1 (highest value — Branch A):

INMET Forecast × Sentinel-2 NDVI → *Optimal soybean and corn spray timing score integrating meteorological window with crop canopy state across the 7 target states.*

NDVI below 0.6 (stressed soybean or corn canopy) combined with a forecast spray window of >4 hours of favorable conditions (wind <10 km/h, humidity 50–80%, no rain <48h) produces a high-confidence APTO spray recommendation on Screen 1. This combination leverages two existing, high-quality sources with a straightforward spatial join (INMET station centroid to field polygon centroid).

Combination 2 (highest value — Branch B):

FieldView planting_summary × MapBiomas LULC archive → *Soil maturity-adjusted soybean and corn planting density recommendation for the 7 target states.*

Combining the FieldView Agronomic Fitness Index (pH fit + density fit + drainage fit) with the MapBiomas *consecutive_crop_years* counter adds a temporal soil stability dimension to the APTO score: a field with 10+ years of continuous crop history in MapBiomas and a Fitness Index ≥ 2 receives an Optimistic tier; the same Fitness Index on a field converted within the last 3 years is downgraded to Baseline, reflecting soil organic matter not yet at equilibrium. This is the most impactful single enhancement to the Branch B plantability model.

SECTION 6

Integrated Data Landscape Visualization

Table 2. Integrated data landscape matrix — 9 sources assessed by structural type, overall quality, decision branch, risk level, and strategic integration priority for Sprints 3–5.

SOURCE	TYPE	QUALITY	BRANCH	RISK LEVEL	STRATEGIC PRIORITY
INMET Historical	Structured	Medium	A	Medium	High
INMET Forecast API	Structured	Medium	A	Medium	Critical
NOAA / GFS	Structured	High	A	Low	High
Sentinel-2	Semi-structured	Medium	BOTH	Medium	Critical
Landsat 8/9	Semi-structured	High	A	Low	High
CEPEA / ESALQ	Structured	Medium	A	Medium	Medium
CBOT / CME Futures	Structured	High	A	Low	Medium
Bayer FieldView	Structured	Medium	B	High	Critical
MapBiomas (Dark)	Semi-structured	High	B	Low	High

Across the 7 target states (MT, PR, MS, RS, GO, MG, SP) and for soybeans and corn specifically, the data ecosystem is structurally asymmetric: Branch A (Spraying Window) is well-served by four high-quality public sources (GFS, Landsat, CBOT, MapBiomass) and two medium-quality sources that can be hardened in Sprint 3 (INMET, Sentinel), while Branch B (Plantability) depends on a single internal dataset with significant completeness gaps. The most critical gap in the ecosystem is the **absence of an external geospatial soil context layer** for Branch B — MapBiomass directly fills this gap and is recommended as the first dark data integration. The recommended sequencing for Sprint 3 data integration is: (1) INMET forecast + NOAA/GFS (define harmonization rule and build unified weather endpoint); (2) Sentinel-2 NDVI pipeline with cloud compositing; (3) MapBiomass centroid extraction for FieldView soybean and corn field polygons within the 7 target states; (4) CEPEA/CBOT price ETL with freshness monitoring; (5) Landsat baseline archive (lower urgency, feeds Screen 2 PREMISSAR scenario only).

SECTION 7 ENHANCEMENT

Shannon Entropy Analysis

Shannon entropy (H, in bits) was computed on the *planting_summary_brazil.csv* dataset (319,788 rows × 29 columns) using `scipy.stats.entropy` with base 2. The six columns subject to TAPI confidentiality restrictions (*field_uuid*, *climate_account_id*, *field_name*, *seed_product_key*, *planting_machine_name*, *concrete_growing_season_name*) were excluded. High H indicates high information diversity (advantageous for classification models); low H indicates concentrated distributions (potential quality issue or low discriminative value). All variables with $H > 2.0$ bits are reported below.

Table 3. Shannon entropy analysis — FieldView *planting_summary_brazil.csv* (319,788 rows × 29 columns). TAPI-restricted columns excluded. Only variables with $H > 2.0$ bits are reported. Color coding: very high H · high H · medium H.

VARIABLE	H (BITS)	UNIQUE VALUES	NULL %	INTERPRETATION	RECOMMENDATION
<i>planted_area_ha</i>	18.278	318,352	0.0%	Near-unique continuous values (one distinct value per row). Expected for a precise area measurement. High diversity is good for regression models but the distribution is heavily right-skewed.	Transform — bin into area classes (e.g., <10 ha, 10–50 ha, 50–200 ha, >200 ha) for categorical tier models.
<i>seed_actual_population_per_hectare</i>	17.953	303,554	0.0%	Near-unique continuous density values. High entropy is expected; bimodal structure (corn	Transform — always stratify by <i>crop_name</i> and filter to

VARIABLE	H (BITS)	UNIQUE VALUES	NULL %	INTERPRETATION	RECOMMENDATION
				≈70k vs. soybeans ≈300k seeds/ha) requires crop-level stratification before any aggregate analysis.	<i>seed_actual_population_uom</i> == 'seeds/ha' before use.
<i>seed_actual_population</i>	17.953	303,554	0.0%	Identical to the per-hectare column pattern; this is the total population before area normalization. Redundant with <i>seed_actual_population_per_hectare</i> for most analyses.	Deprioritize — use the per-hectare version; retain for audit and area consistency checks only.
<i>field_area</i>	13.924	22,604	0.0%	22,604 distinct field sizes indicate precise field-level area measurement. Supports field-specific yield-per-area normalization.	Transform — bin into percentile-based area tiers when used as a categorical feature; retain raw for area-weighted aggregations.
<i>seed_product_name</i>	11.251	16,878	24.0%	16,878 distinct product names; 24% null rate is significant. High diversity reflects the breadth of Bayer's seed catalog but impedes direct grouping without a product taxonomy.	Retain — but implement a product-family normalization layer; the 24% null rate requires imputation or explicit null-class treatment before any seed product analysis.
<i>planting_date</i>	8.487	653	0.0%	653 distinct dates over the observation window; temporally rich. This is the primary variable for planting-window timing analysis and season-length modeling.	Retain — extract day-of-year, growing-degree-days offset, and calendar week as derived features for the plantability scheduler.
<i>city</i>	8.319	701	2.9%	701 distinct cities; good geographic resolution. 2.9% null rate is acceptable. However, city names have free-text noise; the normalized slug form (available via <i>python-slugify</i> in the pipeline) should be used for aggregation.	Retain — aggregate to <i>state</i> level for most analyses; use normalized city slug for finer geographic segmentation.
<i>planting_model</i>	6.484	95	0.0%	95 planting equipment models. Good diversity for equipment-level analysis.	

VARIABLE	H (BITS)	UNIQUE VALUES	NULL %	INTERPRETATION	RECOMMENDATION
					Retain — useful for equipment performance benchmarking in the planter calibration KPI card.
<i>dso_microregional</i>	5.690	60	3.0%	60 Bayer commercial micro-regions; finer than state but coarser than city. Useful for Bayer-aligned geographic segmentation.	Retain — use as the primary geographic grouping for agronomic recommendation calibration aligned with Bayer's commercial structure.
<i>planting_series</i>	3.840	24	0.0%	24 equipment series; superseded by <i>planting_maker</i> + <i>planting_model</i> for most purposes.	Deprioritize — retain as a supplementary equipment attribute but do not use as a primary grouping variable.
<i>state</i>	3.192	15	2.9%	15 Brazilian states in the full dataset. The 7 target states (MT, PR, MS, RS, GO, MG, SP) dominate by record count and define this simulator's geographic scope — consistent with their combined share of Brazilian soybean and corn production. Records from states outside this scope are filtered at ingestion.	Retain — primary geographic stratification variable; filter to the 7 target states at ingestion. Use for all state-level soybean and corn recommendation calibration and reporting.
<i>soil_texture</i>	2.973	10	0.0%	10 texture classes (argilosa, franco-argilosa, etc.); reasonable categorical diversity. Directly relevant to drainage and water retention in the Agronomic Fitness Index.	Retain — integrate as a component of the soil fitness score alongside pH and drainage class.
<i>dso_region_cp</i>	2.475	6	3.0%	6 Crop Protection commercial regions; coarser than DSO microregional but aligned with spray product segmentation.	Retain — use specifically for Branch A spray recommendation calibration where Crop Protection regional data is the appropriate granularity.
<i>planting_maker</i>	2.454	10	0.0%		

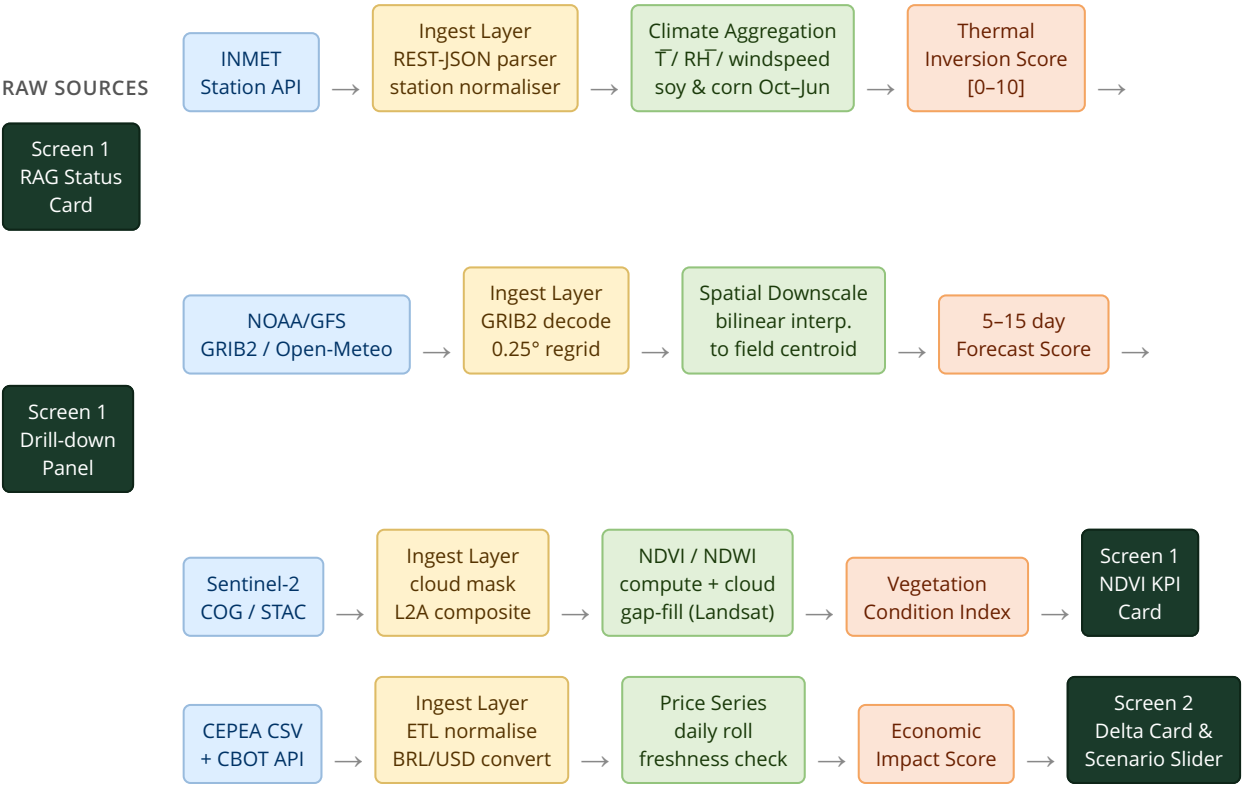
VARIABLE	H (BITS)	UNIQUE VALUES	NULL %	INTERPRETATION	RECOMMENDATION
				10 equipment manufacturers; adequate diversity for equipment-brand segmentation in the planter calibration KPI.	Retain — primary equipment grouping variable; cross-tabulate with <i>avg_operation_speed_kmh</i> (from operations dataset) for speed-by-maker benchmarking.
<i>dso_region_seeds</i>	2.424	7	10.7%	7 seeds commercial regions; 10.7% null rate is the most significant geographic missingness in the dataset.	Retain with caution — 10.7% nulls must be explicitly handled; do not use as a mandatory grouping variable in production models without null-class treatment.
<i>soil_ph_max</i>	2.086	5	0.0%	Quality alert: Only 5 distinct values for a supposedly continuous pH measurement strongly indicates that pH data is coarsely binned (e.g., pH intervals: <5.0, 5.0–5.5, 5.5–6.0, 6.0–6.5, >6.5) rather than measured precisely. This severely limits the discriminative power of pH-based fitness scoring.	Transform — treat as an ordinal categorical variable; replace the continuous pH fitness band comparison in <code>src/scenarios.py</code> with an ordinal tier lookup; document this as a data quality limitation in all Branch B outputs.
<i>soil_order</i>	2.051	8	0.0%	8 SiBCS soil orders (Latosolo, Argissolo, etc.); good agronomic diversity reflecting the varied Cerrado pedology.	Retain — use as a soil fertility proxy in conjunction with pH and drainage for the Agronomic Fitness Index enhancement.

Key finding — Soil pH coarse binning: The H=1.538 bits for *soil_ph_min* (4 unique values) and H=2.086 bits for *soil_ph_max* (5 unique values) are the most actionable quality signals in this dataset. If soil pH were continuously measured, H would be expected to be >8 bits (hundreds of distinct values). The low H confirms the A1 EDA observation that pH data represents binned categorical ranges. Any agronomic threshold comparison (e.g., "soybean optimal pH 6.0–7.0") applied to these bins will have reduced precision, and the Cramér's $V = 0.0035$ (NEGLIGIBLE) between the Fitness Index and yield tiers in the A1 EDA is consistent with this low-resolution pH encoding.

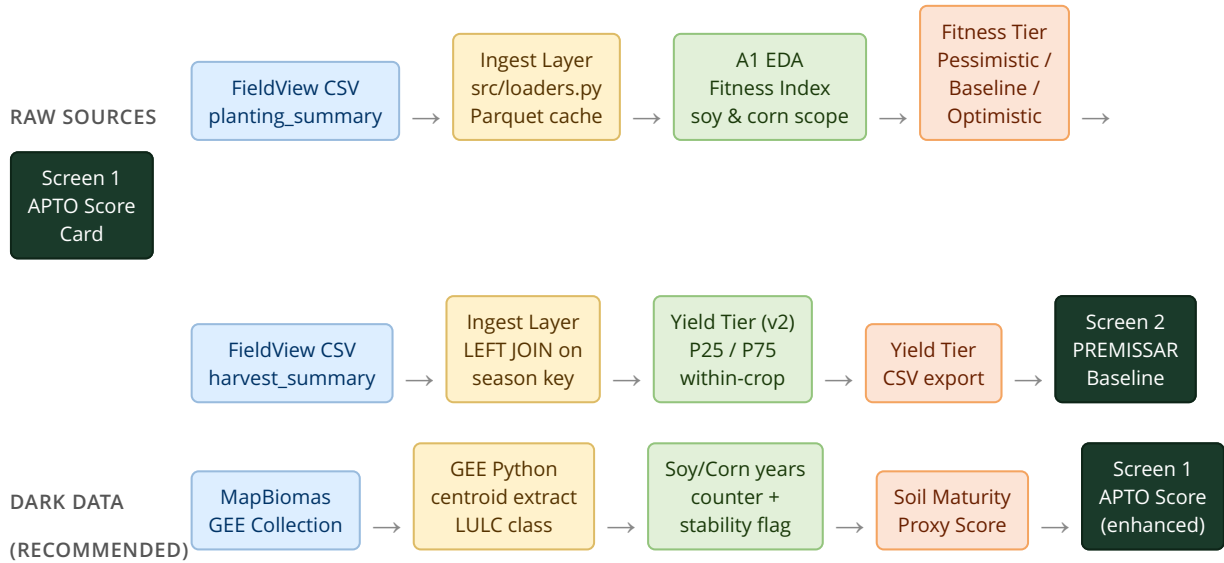
Data Lineage Diagram

The diagram below maps both simulator pipelines from raw source to Screen output. Each node represents a distinct processing stage; arrows indicate data flow direction.

Branch A — Spraying Window Pipeline



Branch B — Plantability Pipeline



SECTION 9 ENHANCEMENT

Data Governance Recommendations

The two highest-risk sources identified in Section 5 are **Bayer FieldView** (TAPI confidentiality + completeness gaps) and **INMET Historical** (2022 station-renaming discontinuity + no versioned API). Governance frameworks for each are proposed below.

9.1 Bayer FieldView Dataset

GOVERNANCE DIMENSION	RECOMMENDATION
Ownership	Assign one named team member as <i>FieldView Data Steward</i> (role, not person-bound). Responsibilities: maintain the CHECKSUMS.txt gate, liaise with Bayer project leader (William Alves) for data refresh, and review TAPI restrictions before each new use of the data.
Update cadence	Data refresh is annual (batch export). The Steward requests updated CSVs from Bayer at least 4 weeks before any new sprint that modifies Branch B models. A new git tag (<code>data-fieldview-YYYY-MM</code>) must be created for each refresh and documented in <code>data/raw/CHECKSUMS.txt</code> .

GOVERNANCE DIMENSION	RECOMMENDATION
Version control	Raw CSVs are never committed to git (gitignored). SHA-256 checksums are committed in <code>data/raw/CHECKSUMS.txt</code> and verified at every notebook execution via the <code>make reproduce</code> target. The Parquet cache must be regenerated whenever the SHA-256 changes.
Access control	Raw CSVs are stored only in the <code>paths.yml.local</code> -configured directory on each team member's local machine. They must not be uploaded to shared drives, cloud storage, or any repository. Encrypted storage (macOS FileVault or equivalent) is required on team machines holding the raw data.
Escalation path	If any quality issue (e.g., unexpected null surge, new column, schema change) is detected by the SHA-256 gate or the automated quality check in <code>src/quality.py</code> , the Steward notifies William Alves within 24 hours and opens a project issue tagged <code>data-quality:fieldview</code> before any further analysis proceeds on the affected dataset.

9.2 INMET Historical Dataset

GOVERNANCE DIMENSION	RECOMMENDATION
Ownership	Assign one named team member as <i>INMET Data Steward</i> . Responsibilities: monitor the INMET API changelog (api.inmet.gov.br/docs), manage the station-code translation table, and run the monthly ETL health check.
Update cadence	Historical data is fetched once per sprint (monthly) and cached as Parquet. The station-code translation table (<code>data/interim/inmet_station_map.parquet</code>) is versioned separately with its own SHA-256. Any detected change in the station-code mapping triggers a full historical reprocessing before the next model run.
Version control	Tag each historical fetch with the API access date: <code>data-inmet-historical-YYYY-MM-DD</code> . Store the API response schema version (obtainable from the INMET API root endpoint <code>/version</code>) in the tag annotation. Schema version mismatch triggers a mandatory ETL review.
Access control	INMET API tokens (free registration) are stored in environment variables (<code>INMET_API_TOKEN</code> in <code>.env</code>) and never committed to git (confirmed in <code>.gitignore</code>). Token rotation is the Steward's responsibility on a 90-day cycle.
Escalation path	Monthly ETL health check runs the station-code translation on the 10 most-recent API responses. If >5% of station codes are unresolved (indicating a new renaming event), the Steward halts Branch A data ingestion, updates the translation table, and notifies the team before the next pipeline run. Fallback: use NOAA/GFS exclusively for any date range where INMET data is under review.